Statistics for Economics and Management

Practical class #1 – Review on random variables

Francisca Ramirez

maria.ramirez@novasbe.pt

Office Hours: Wednesdays 18:30
(by appointment | in-person in room B128 or online)

Some information about the course

- **My e-mail:** maria.ramirez@novasbe.pt
  - For letting me know you want to attend my **Office Hour** – please send an e-mail **until the day before**
  - And for posing any **questions** about the course/materials that can be easily answered by e-mail (for more complex doubts, please attend the Office Hours).
- **Moodle:**
  - Enrollment password: **Gauss**
  - You will find all course materials and important information (including the Syllabus) on Moodle
  - Please also use the **Q&A forum** for doubts: your doubts are likely to be someone else's doubts too!
- **Assessment**:
  - Midterm Exam: 35% – March 14th (Saturday) at 17:30
  - Final Exam (Regular season): 65% – June 6th (Saturday) at 17:30
  - Resit Exam: 100% – June 25th (Saturday) at 18:30
  - (For students enrolled in grade improvement, the final exam (Regular or Resit) counts 100%)

Please bring your laptops to class next week. We will solve two exercises on Excel.

Before the exercises...

Let’s do a short theoretical recap of what you need to know!

Disclaimer:

The practical sessions only focus on some key contents of the course. For a complete and more thorough analysis of the course’s topics, please attend the theoretical classes and check out the Professor’s materials on Moodle!

Random variables

A **random variable** is a function that associates a real number to an event in an event space.

- **Discrete** random variables are those that take a countable number of values. This means that the possible values that the random variable may take form a discrete set.

  A discrete random variable is fully characterized by its **probability distribution function**:

  $f(x) = P(X = x)$, which in turn defines the **cumulative probability distribution**: $\mathrm{F}(x) = P(X \leq x)$.

- **Continuous** random variables may take any value in a given interval. This means that the possible values that the random variable may take form a continuous set.

  A continuous random variable is fully characterized by its **probability density function**:

  $f(x) \neq P(X = x)$ [since $P(X = x) = 0$ for any continuous random variable], which in turn defines the **cumulative probability distribution**: $\mathrm{F}(x) = P(X \leq x)$.

Random variables

Expected Value - $E(X) = \mu$

The expected value of a random variable is a measure of **central location**:

- $E(X) = \sum_x x \cdot f(x)$ for discrete variables
- $E(X) = \int_x x \cdot f(x)dx$ for continuous variables.

Properties:

- $E(a) = a$ for any constant a.
- $E(aX) = aE(X)$ for any constant a and any random variable X.
- $E(aX + bY) = aE(X) + bE(Y)$ for any constants a and b, and any random variables X and Y.

Random variables

Variance - $\mathrm{Var}(X) = \sigma^2$

The variance of a random variable is a measure of **dispersion**: $\mathrm{Var}(X) = E(X - \mu)^2$

- $Var(X) = \sum_x x^2 \cdot f(x) - \mu^2$ for discrete variables.

- $Var(X) = \int_x x^2 \cdot f(x)dx - \mu^2$ for continuous variables.

Properties:

- $Var(a) = 0$ for any constant a.

- $Var(aX) = a^2 Var(X)$ for any constant a and any random variable X.

- $Var(aX + bY) = a^2 Var(X) + b^2 Var(Y) + 2abCov(X, Y)$ for any constants a and b, and any random variables X and Y.

Random variables

Covariance - $Cov(X,Y) = \sigma_{X,Y} = E[(X-\mu_X)(Y-\mu_Y)]$ *or* $Cov(X,Y) = E(\mathrm{X}\cdot\mathrm{Y}) - \mu_X \cdot \mu_Y$

The covariance measures the **direction** of the **linear relationship** between two random variables. The absolute value of the covariance itself has no meaning, we may only interpret its sign:

- $\mathrm{Cov}(X,Y) > 0$: values above/below the mean in one variable are usually associated with values above/below the mean in the other variable.
- $\mathrm{Cov}(X,Y) < 0$: values above/below the mean in one variable are usually associated with values below/above the mean in the other variable.
- If X and Y are independent, then $\mathrm{Cov}(X,Y) = 0$. **But** if we have that $\mathrm{Cov}(X,Y) = 0$ then X and Y are **not necessarily** independent! - the covariance detects **linear** relationships, but variables may be related in many other ways (quadratic, exponential...). So: $\mathrm{Cov}(X,Y) = 0 \not\Rightarrow X$ and Y are independent.

Random variables

Correlation - $Corr(X,Y) = \rho_{X,Y} = \frac{Cov(X,Y)}{\sqrt{Var(X)} \cdot \sqrt{Var(Y)}} = \frac{\sigma_{X,Y}}{\sigma_X \cdot \sigma_Y}$

The correlation measures the **direction** and **strength** of the **linear relationship** between two random variables:

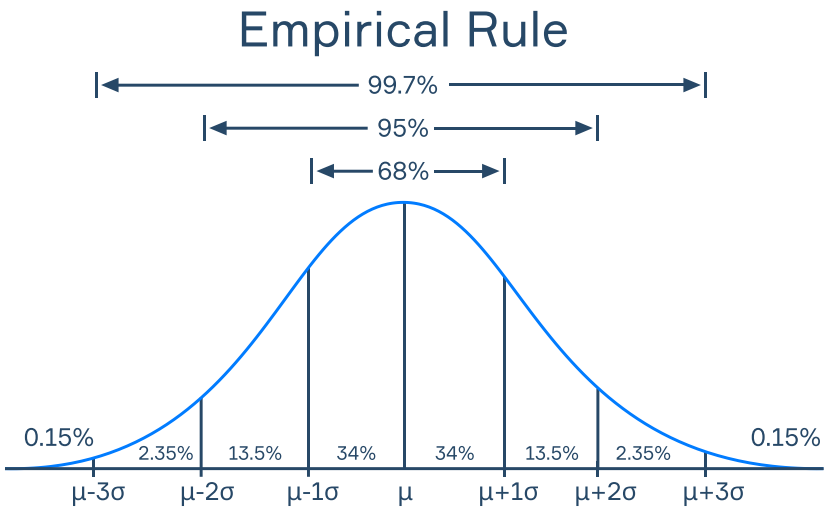
- $0 < \mathrm{Corr}(X,Y) \leq 1$: values above/below the mean in one variable are more likely to be associated with values above/below the mean in the other variable.
- $-1 \leq \mathrm{Corr}(X,Y) < 0$: values above/below the mean in one variable are more likely to be associated with values below/above the mean in the other variable.
- $\mathrm{Corr}(X,Y) = 0$: no **linear** relationship between X and Y.
- Note that we can rewrite $\boldsymbol{Cov(X,Y) = \rho_{X,Y} \cdot \sigma_X \cdot \sigma_Y}$

Normal distribution: $X \sim N(\mu, \sigma^2)$

- Centered around its mean (μ), and **symmetric**.
- The area under the curve is equal to 1.
- The **standard normal distribution** has mean 0 and variance equal to 1: $Z \sim N(\mu = 0, \sigma^2 = 1)$.
- In order to compute probabilities, we must **standardize**:

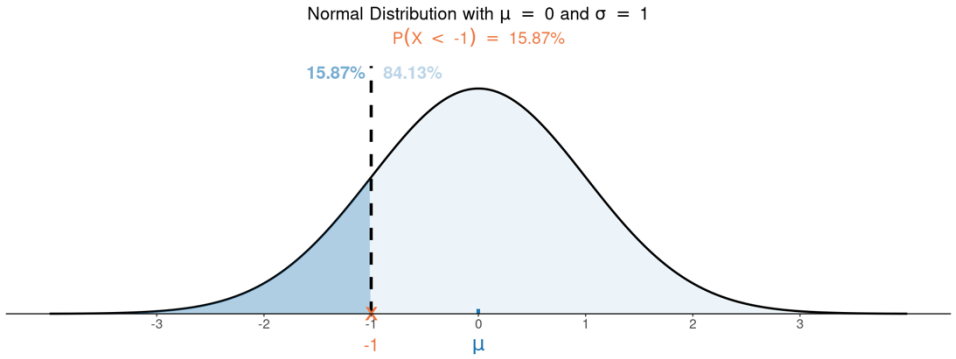
  $P(X < x) = P(\frac{X-\mu}{\sigma} < \frac{x-\mu}{\sigma}) = P(Z < \frac{x-\mu}{\sigma}) = P(Z < z)$

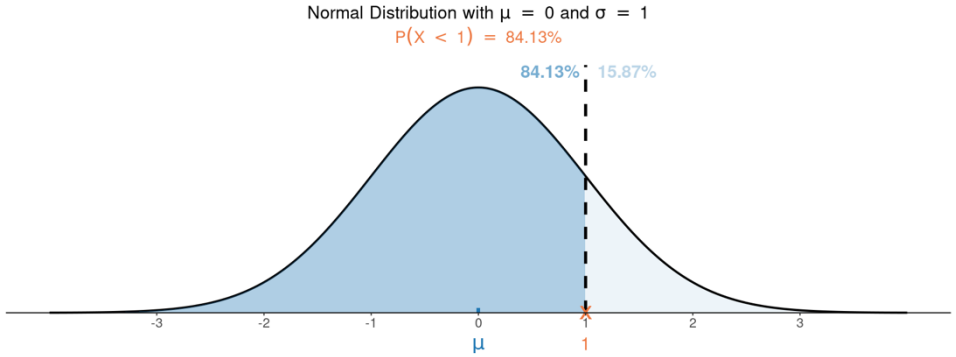
  which (for $\boldsymbol{z > 0}$) we can check in the Normal distribution statistical table (available in Moodle).
- If $\boldsymbol{z < 0}$, we take advantage of symmetry and write $P(Z < z) = 1 - \mathrm{P}(Z < -\mathrm{z})$.
- Remember that $\boldsymbol{P(Z > z) = 1 - \mathrm{P}(Z < z)}$.



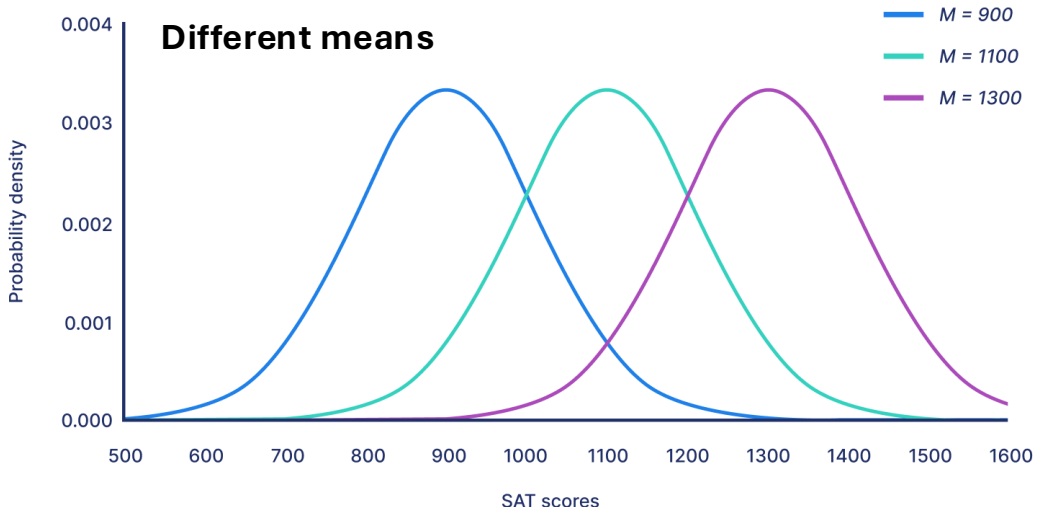
Normal distribution: $X \sim N(\mu, \sigma^2)$

Symmetry

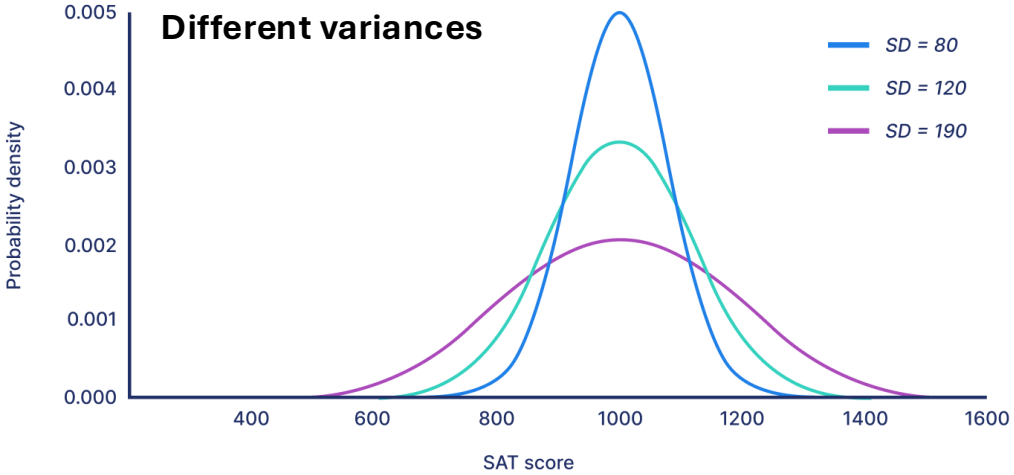




Different means



Different variances



Workbook, exercise 1.22

In the course of an investigation regarding a complaint about the warranty period of the ROADSTER brand of automobile tires, a consumer safety commission is analyzing the quality of the production process. The manufacturer gives a 45 thousand km warranty on these tires. However, it also knows that due to variability in the production process and tire usage, the duration of a tire under regular usage conditions is random with a normal distribution with mean 50 thousand km and standard deviation 5 thousand km.

(a) What is the probability that a random tire lasts less than 50 thousand km (you don't need to check the Normal distribution table to answer this)?

(b) What is the probability that a randomly selected tire will last less than the 45 thousand km given by the warranty?

(c) What is the probability that a random tire lasts between 49 and 51 thousand km?

(d) There is a 5% probability that the duration of a tire is extremely low. What does extremely low mean?

Workbook, exercise 1.28

Suppose that you are responsible for managing the sales department of a computer company with two shops in Lisbon. In the following month, these two stores will put on sale the new UniPad tablet. You must decide today how many units should be purchased from the manufacturer in order to jointly meet the demand in the two shops in Lisbon in July. The demands of UniPad tablets in the two stores in the following month are uncertain and may be considered as random variables. The demand in the first store can be described by a normal distribution with mean 40 and standard deviation 4. The demand in the second store can be described by a normal distribution with mean 60 and standard deviation 3. Assume these two random variables have a joint normal distribution.

(a) What is the probability that the demand for UniPad tablets in the first store exceeds 46 units next month?

(b) If the correlation between demands in the two stores equals 0.5, what is the variance of the total demand for the UniPad tablets in the two stores (given by the sum of the demands in the first and second stores) next month?

Workbook, exercise 1.28

(c) If the total demand in the two stores is below the number of units purchased from the manufacturer, there will be a loss. On the other hand, if the total demand exceeds the number of units purchased to the manufacture, then it will not be possible to satisfy all the demand and some of your clients will be unhappy. Answer the following two questions:

i) Suppose it is possible to assume that in July the demands for UniPad tablets in the two stores are **independent** of each other. How many units should you purchase from the manufacturer so that you can satisfy total demand (the sum of the demands in the first and second stores) with a probability of 85%?

ii) If the demands of UniPads in the two stores are **correlated**, what is the value of the correlation which would require a maximum purchase from the manufacturer to ensure it is possible to meet total demand with a probability of 85%?

Workbook, exercise 1.13

In the final two weeks of a promotion campaign, a firm must decide who will be the operational supervisor in each of these two weeks. There are two candidates: Carla and Maria. On average, both Carla and Maria are expected to achieve the same sales per week. Assume that the standard deviation of weekly sales is also the same for both. If Carla supervises in one week, and Maria supervises in the other week, sales in each week are independent. However, if the same person supervises in both weeks, sales in each week are positively correlated since if one of the supervisors performs above (below) average in the first week, it is more likely that he will also perform above (below) average in the second week. This joint decision for the two weeks has to be made today, in advance of those two weeks, and cannot be changed afterwards.

(a) In terms of expected total sales in the two weeks, what is the best choice of supervisor for each week? Statistically justify your answer.

(b) If the firm is risk averse, what is the best choice of supervisors in each week? Provide the statistical details to justify your answer.